N83 Q0: reconciling A (rho faster) and B (IC faster)

Status (2026-10-14): source audit done; protocol frozen; the Tier 1 run starts on push of branch `claude/n83-reconcile`. No timing result exists yet. This directory makes no performance claim. Results will land as `run_github_<id>/` with `result.json` and `result_table.md`, and this README will be updated with them. The workflow's path filter ignores those files.

Two measurements of the same point disagree in direction:

	A	B
Where	crypto `research/n83_evaluation_20261012/`	cryptanalysis `experiments/koblitz-single-target-n83-tie` (run 38523707244)
Crypto source	`a8194d61`	`770f6a69`
Rho	`StrongRhoG<WideGf2,u64>` default (32 lanes, DP 4, escape off), seeds 202610061..65	`koblitz_rho_batch_ks_strong_online` rung 3 (32 lanes, DP 4), seed 831000
IC	`koblitz_orbit_dlp_fast`, fresh in-process rank (seeds 21..25)	`koblitz_orbit_dlp_fast_online` + `ic_log_cache.patch`, cached factor-log table
Host	5-vCPU KVM guest, one pinned CPU	GitHub ubuntu-24.04 runner, one pinned CPU
Q0 result	IC/rho online 1.382 [1.036, 1.882] (5 pairs)	rho/IC online 1.779 [1.775, 1.783] (8 pairs, one rho walk)
Reading	rho online faster	IC online faster

Point Q0 = (265589834553716618053296 0, 388152630799167326529261 5) on `EC1N83Ce1h35c7ced6b497` (y^2 + xy = x^3 + x^2 + 1 over GF(2)[z]/($z^{83+z}7+z^{4+z}2+1$), r = 8569786107849059), retained K600 base `d4a4da26...a9f7`.

Source audit (before any timing)

SOURCE_AUDIT.md (PDF). Both IC runs did identical work on Q0: 8,845,441 probes and the same relation, at every rank seed, cached or fresh. So the gap splits exactly into rho steps x ns per step over IC probes x ns per probe:

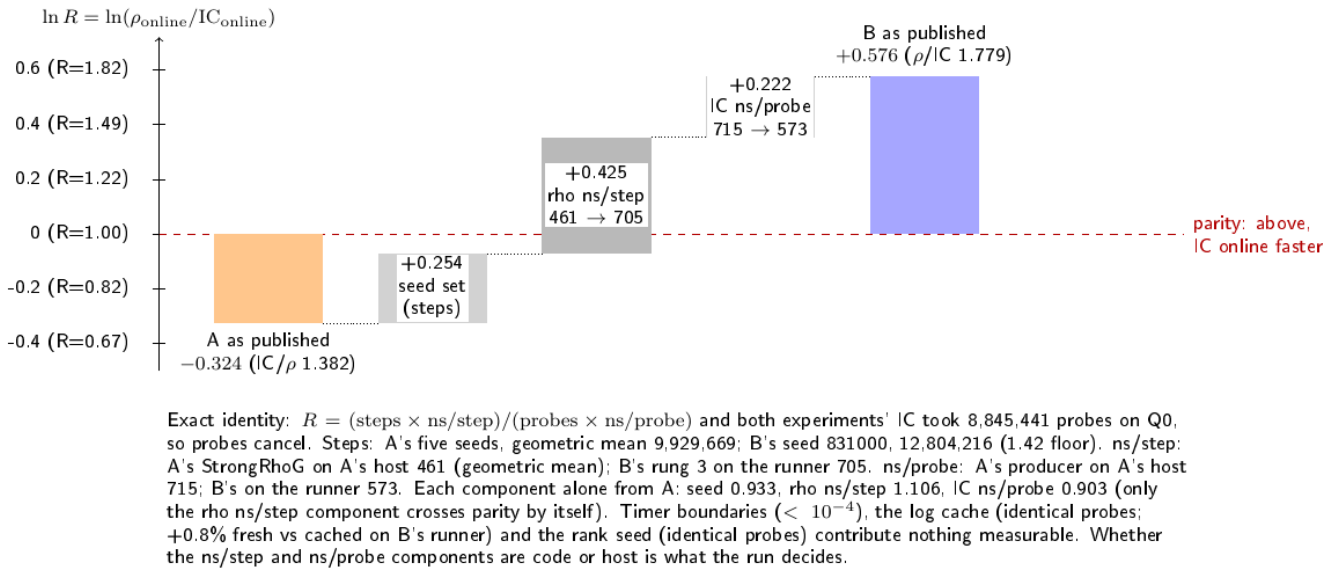


Figure 1: Exact decomposition of the published gap

Figure 1. The published A to B gap in ln(rho/IC) on Q0, split into its three exact components from the raw receipts (source, vector). These are not one-host measurements.

- **Explained without a run:** the two harnesses' timer boundaries cover the same operations (difference under 1e-4 in ln R). The log cache moves no work (identical probes; +0.8% fresh against cached on B's runner). The rank seed is irrelevant online. A's median agrees with A's geometric mean. B's seed made Q0 a long walk: 1.42 times the floor, against a geometric mean of 1.10 over A's five seeds. That gives +0.254 in ln R, which cannot flip A's direction alone (0.933).
- **Not explained without a run:** rho ns per step is 461 for A's StrongRhoG on A's host and 705 for B's rung 3 on the runner (+0.425, enough to flip alone: 1.106). IC ns per probe is 715 for A's producer on A's host and 573 for B's on the runner (+0.222). Each could be code or host: both hosts report an AMD EPYC 9V74 and rustc 1.99.0, but they are different VMs.

Frozen design (protocol.json, schedule.json)

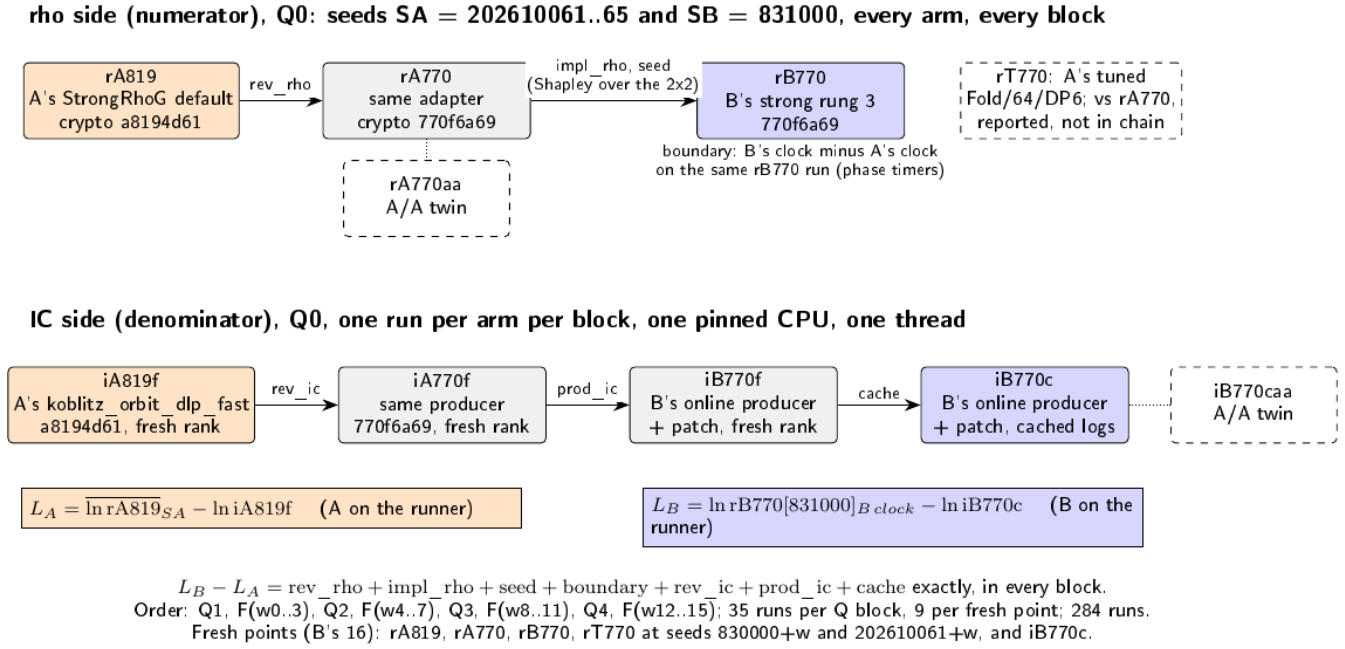


Figure 2: Arms and the seven-term chain

Figure 2. Arms and the exact chain from A's configuration to B's (source, vector).

Arm	Code	Revision	Role
rA819	A's `n83_strong_point_eval` (StrongRhoG default)	a8194d61	A's exact rho
rA770	same adapter	770f6a69	rho revision
rA770aa	same as rA770	770f6a69	A/A control
rB770	rung 3	770f6a69	B's exact rho
rT770	A's tuned worker, Fold/64/DP6	770f6a69	reported, not in the chain
iA819f	A's `koblitz_orbit_dlp_fast`, fresh sequential rank	a8194d61	A's exact IC
iA770f	same producer	770f6a69	IC revision
iB770f	B's online producer + patch, fresh sequential rank	770f6a69	IC producer
iB770c	same, cached table (`KIC_LOG_TABLE_IN`)	770f6a69	B's exact IC
iB770caa	same as iB770c	770f6a69	A/A control

- **Points:** Q0, plus B's 16 fresh public points (B's `inputs/targets.json`, hash-bound).
- **Q0 panel:** 4 complete blocks. Each block runs every rho arm at all six seeds (A's 202610061..65 and B's 831000) and every IC arm once: 35 runs, in a frozen pseudo-random order. Fresh-rank arms use rank seed 20 + block.
- **Fresh panel:** per point, rA819, rA770, rB770 and rT770 at seeds 830000 + w (B's rule) and 202610061 + w (A's family), and one cached IC run. That is 9 runs per point.
- **Order:** Q1, F(w0-3), Q2, F(w4-7), Q3, F(w8-11), Q4, F(w12-15); 284 runs in all. One process at a time on one pinned CPU, in one job.
- **Factorial:** every arm runs in every block, so every rho arm x seed x IC arm combination is a within-block paired ratio. Rho implementation (3 levels) x seed set (2) is fully crossed. Revision is nested in A's code, and the log source in B's IC producer, because the other code does not exist at n=83 at a8194d61 (SOURCE_AUDIT section 5).
- **Statistic:** natural logs. Per block, `L_A` = mean over A's seeds of ln rA819 minus ln iA819f; `L_B` = ln rB770[831000] (B's clock) minus ln iB770c. Then `L_B - L_A` = rev_rho + impl_rho + seed + boundary + rev_ic
  - prod_ic + cache, exactly. impl_rho and seed are Shapley values over the 2x2. The bootstrap resamples blocks (10,000 draws; 95% intervals, and Bonferroni 99.29% for m = 7). On the fresh points: each rho factor per point, bootstrapped over points, and B's section 23 statistic per rho arm.
- **Decision rule:** a factor is *sufficient to flip* if A reproduces on the runner (L_A upper bound < 0) and L_A + D_f has a Bonferroni lower bound > 0. It *contributes* if its Bonferroni interval excludes 0 and $|D_f| >= 0.05$. It is *negligible* if its 95% interval lies inside [-0.05, 0.05]. The responsible factors are every sufficient one, or else the smallest greedy set of contributing ones that flips L_A. If A's configuration already reads IC-faster on the runner, the responsible factor is `host_or_instance`.
- **Controls:** two A/A twins (rho on all six seeds, and cached IC). Each passes if $|\text{mean ln}| <= 0.03$ and its 95% interval lies inside [-0.05, 0.05]. A positive control: the instrument must recover, within 0.10, the exact ln step ratio 0.978 between rA770's seeds 202610064 and
202610063. Identity checks: one scalar per point, equal to Q0's sidecar and replayed in `oracle.py`; identical steps per seed across rA819, rA770 and rA770aa; identical probes across IC arms; every fresh table's logs equal to B's archived table.
- **Validity:** invalid rows (exit, verification, timeout, memory event) make `no_result`. Contention flags from isolated_bench's Tier 1 gate (steal, throttling, major faults) keep the row. The decision is recomputed without flagged blocks, and a different answer reads `sensitive_to_contention`.

Resources and expected runtime

Stage	Estimate	Basis
rustup 1.99.0, two crypto builds, bin tests	15-30 min	local builds 5.6 + 4.7 min at -j3
B's table re-verified in oracle.py	about 10 s	local
Q0 block (3 fresh-rank IC at about 14-18 min each, 2 cached IC, 30 rho)	48-62 min	B runner: fresh B producer 888 s, cached 52 s, rung 3 705 ns per step; A host: A's IC 1,054 s per process
Fresh segment (4 points x 9 runs)	6-8 min	B Panel B: IC median 645 ms online + 46 s setup, rho 6.2 s
Audit	5-15 min	oracle replays and log-table checks
Total	**about 4.1-5.3 h**	job timeout 350 min; a time guard skips whole segments past 325 min

Memory: fresh-rank IC peaks at about 5.8 GB, one process at a time, within 16 GB.

Local validation (correctness only; no timing is reported)

Local validation ran in a shared, contended container. Receipts: smoke/.

- `restore_sources.py`: both snapshots match `protocol.json`. Every file shared with B also matches B's frozen digests, and A's adapter matches A's recorded 82bc92d2.
- Both crates build with `--locked`. `cargo test --release --locked --bins` passes: 12 tests (r819) and 32 tests (r770).

- `run_schedule.py --segments Q1,F1 --one-per-arm` ran all 10 arms on Q0 and the 5 fresh-panel arms on point w00: 15 valid rows. Then `verify_receipt.py --correctness-only` passed every identity check (see the list after this one).
- Earlier direct `arm.py` calls reproduced A's and B's exact step counts on Q0: 10,203,425 at seed 202610061 for rA819 and rA770, and 12,804,216 at seed 831000 for rB770.
- `make_schedule.py --check` passes. `verify_log_table.py` passes on B's archived table. `verify_receipt.py` on synthetic data reproduced the chain identity to rounding, and a deadline-0 run of `run_schedule.py` recorded all 284 runs as `skipped_time_guard`.
- Workflow: YAML parses; `actionlint` 1.7.7 with shellcheck 0.10.0 passes.

The identity checks that passed:

- one scalar per point; Q0's equals the sidecar 3141592653589793; every scalar replays in `oracle.py`;
- steps equal A's receipts (14,229,982 and 13,159,092 at seeds 202610064 and 202610065) and B's (9,447,525 on w00 at seed 830000);
- probes are 8,845,441 for every IC arm on Q0 and 14,455,639 on w00, as in B's Panel B;
- the fresh B-producer table's 600 logs equal B's archived table.

Gaps and limits

- The chain is measured on **one runner**. A residual between A's published L_A and the runner's L_A is the A-host effect, and it cannot be split further here.
- B's IC producer and rung 3 have no n=83 path at a8194d61. Their revision effects are therefore not estimable, and only A's code carries the revision factor.
- IC producer, revision and log-source effects are measured on Q0 only (each fresh-rank arm costs about 15 min). The fresh points carry the rho factors over targets.
- Seed-set effects are fixed effects of these seeds. Q0 intervals cover repeat and timing variation, not seed or target variation.
- Tier 1 instance scope: one pinned exposed CPU on a shared VM. Exclusivity is not certified.
- Canonical graphs checked: B's `figures/prior_ratios.*` and `figures/tier1_result.*`, and `experiments/volcano-ic/fig`. None changes until a result exists. This directory's figures are new.

Files

`protocol.json` (frozen design and digests), `schedule.json` (frozen order), `SOURCE_AUDIT.md`, `restore_sources.py`, `crates/{r770,r819}/`, `arm.py`, `make_schedule.py`, `run_schedule.py`, `verify_log_table.py`, `verify_receipt.py`, `figures/`, `smoke/`, and the workflow `.github/workflows/n83-reconcile.yml`. The workflow reuses from B: `ic_log_cache.patch`, `inputs/targets.json` and the archived factor-log table of run 38523707244.

Reproduce

```
git clone https://github.com/aburan28/crypto /path/crypto
python3 restore_sources.py --crypto-repo /path/crypto
python3 make_schedule.py --check
(cd crates/r819 && cargo build --release --locked); (cd crates/r770 && cargo build --release --locked)
python3 verify_log_table.py --table ../koblitz-single-target-n83-tier1-20261014/tier1_github_38523707244/pr
python3 run_schedule.py --bin-dir $CARGO_TARGET_DIR/release \
  --log-table ../koblitz-single-target-n83-tier1-20261014/tier1_github_38523707244/precompute/factor_logs_n
  --receipt-dir R/panel --cpu 0
python3 verify_receipt.py R/panel
```